⌈Def Let $D \in \mathbb{Z}$ be a square-free integer.
Then, a set $\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$ forms a field, and
$\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$ forms a Ring as a subring of $\mathbb{Q}(\sqrt{D})$.
Further, if $D \equiv 1 \mod 4$, then for $w = \dfrac{1+\sqrt{D}}{2}$,

$$\mathbb{Z}[w] = \{a + bw \mid a, b \in \mathbb{Z}\}$$

is also subring of $\mathbb{Q}(\sqrt{D})$, and contains $\mathbb{Z}[\sqrt{D}]$. Figure out now.
At first, $\mathbb{Q}(\sqrt{D})$ is vector space over $\mathbb{Q}$, with basis $\{1, \sqrt{D}\}$. So, define a Norm as

$$N : \mathbb{Q}(\sqrt{D}) \to \mathbb{Q} : a + b\sqrt{D} \mapsto a^2 - D \cdot b^2$$

Actually, $N(\alpha) = \prod_{\sigma \in Gal(\mathbb{Q}(\sqrt{D}), \mathbb{Q})} \sigma(\alpha)$. So, if $\alpha \neq 0$, $\alpha^{-1} = \dfrac{\overline{\alpha}}{N(\alpha)}$.
In $\mathbb{Q}(\sqrt{D})$, It is clearly abelian group under the addition, and by the fact above,
$\mathbb{Q}(\sqrt{D}) - \{0\}$ is abelian group under the multiplication.
Now, let's to dissect $\mathbb{Z}[\sqrt{D}]$. Clearly $\mathbb{Z}[\sqrt{D}]$ is closed under the substraction.
And, for $a + b\sqrt{D}$, $c + d\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$,

$(a + b\sqrt{D}) \cdot (c + d\sqrt{D}) = ac + bd \cdot D + (ad + bc) \cdot \sqrt{D}$, So closed under the multiplication.

Associative, Commutative, Containing 0 and 1 are clear.
Moreover, for $\mathbb{Z}[w]$, it is clearly group under the addition, and,

$$\left(a + b \cdot \frac{1+\sqrt{D}}{2}\right) \cdot \left(c + d \cdot \frac{1+\sqrt{D}}{2}\right) = ac + bd \cdot \frac{D-1}{4} + (ad + bc + bd) \cdot \frac{1+\sqrt{D}}{2}$$

is contained in $\mathbb{Z}[w]$, since $(D-1)/4 \in \mathbb{Z}$, by $D \equiv 1 \mod 4$.
(More observation: for $a + b\sqrt{D}$ with $b \neq 0$, a minimal Polynomial over the given Ring is

$$(x-a)^2 - D \cdot b^2 = x^2 - 2ax + a^2 - D \cdot b^2$$

which has another root as $a - b\sqrt{D}$.)

「Thm. The field Norm $N: \mathbb{Q}(\sqrt{D}) \mapsto \mathbb{Q}$; $a+b\sqrt{D} \mapsto a^2 - Db^2$ preserves the multiplication.

Proof. Let $a+b\sqrt{D}, c+d\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ be given. Then,

$$\begin{aligned}
N([a+b\sqrt{D}]\cdot[c+d\sqrt{D}]) &= N(ac+bd\cdot D + (ad+bc)\sqrt{D}) \\
&= (ac+bd\cdot D)^2 - D\cdot(ad+bc)^2 \\
&= a^2c^2 + b^2d^2D^2 + 2abcd\cdot D - D\cdot(a^2d^2+b^2c^2+2abcd) \\
&= a^2c^2 + b^2d^2D^2 - D\cdot(a^2d^2+b^2c^2) \\
&= (a^2 - Db^2)\cdot(c^2 - Dd^2) \\
&= N(a+b\sqrt{D})\cdot N(c+d\sqrt{D}).
\end{aligned}$$

「Thm. In $\mathbb{Z}[\sqrt{D}]$ or $\mathbb{Z}[\omega]$, an element α is a unit iff $N(\alpha) = \pm 1$.

Proof. If α is unit, then $N(\alpha\cdot\alpha^{-1}) = N(\alpha)\cdot N(\alpha^{-1}) = 1$.

But, since $N(\alpha), N(\alpha^{-1}) \in \mathbb{Z}$, both are 1 or both are -1.

Conversely, if $N(\alpha) = \pm 1$, $\dfrac{\alpha}{N(\alpha)}$ is inverse of α.